

Database Schema Documentation

Labor Market Monitor Data Warehouse

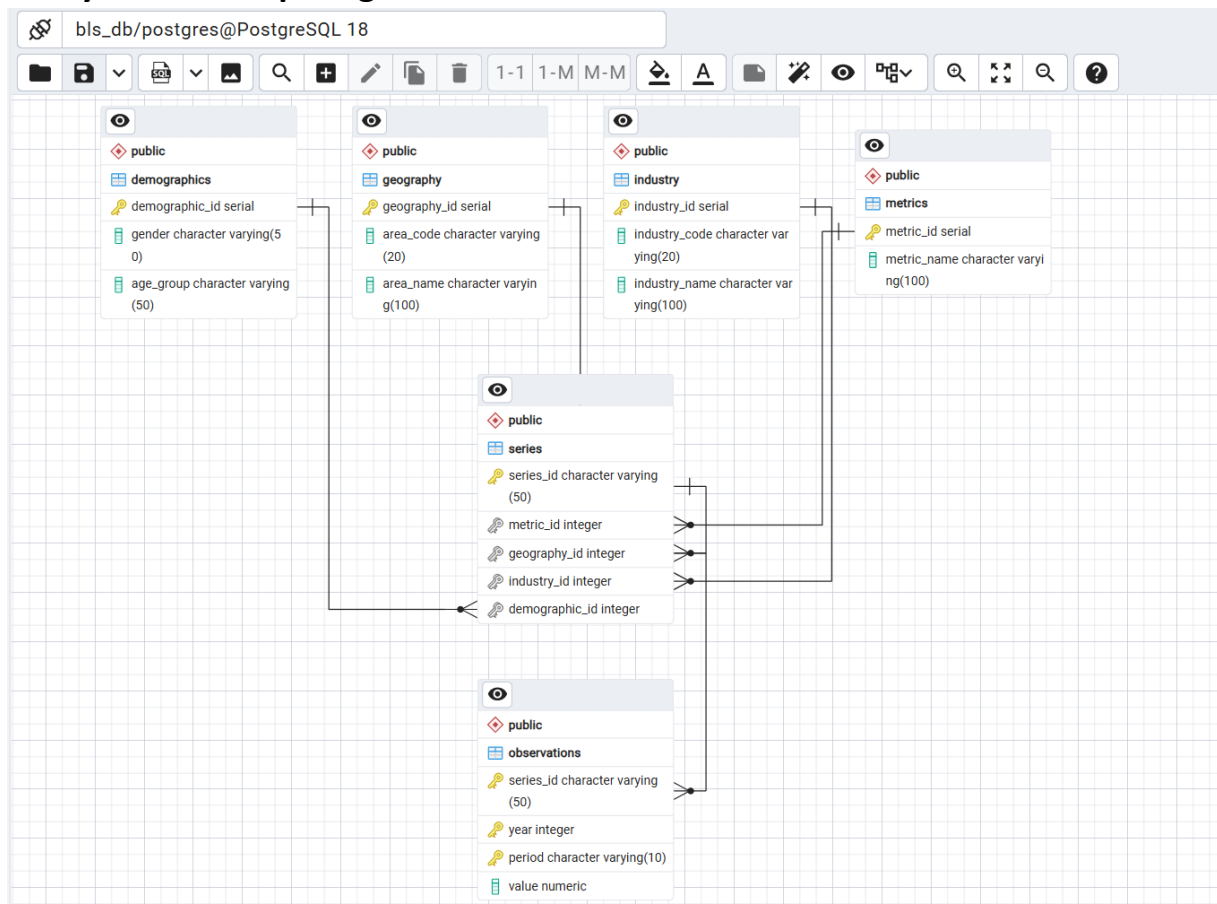
This database stores labor market and industry employment data retrieved from the U.S. Bureau of Labor Statistics Public API V2 and loaded into a PostgreSQL database using Python, Pandas, and SQLAlchemy as part of an ETL pipeline for interactive dashboard analytics.

Database Overview

The database contains six tables:

- metrics
- geography
- industry
- demographics
- series
- observations

Entity Relationship Diagram



Entity Relationship Summary

Table	Purpose
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metrics	Stores labor market metric definitions
geography	Stores geographic region information
industry	Stores industry classification information
demographics	Stores demographic categories for future expansion
series	Stores BLS series metadata and dimensional relationship
observations	Stores monthly labor market observation values

Table Documentation

1. metrics

Purpose

Stores labor market metric definitions used throughout the dashboard.

Examples:

- unemployment_rate
- employment_level
- labor_force

Primary Key

metric_id

Relationships

One metric can relate to many series records.

Referenced by:

series.metric_id

Table Structure

Column Name	Data Type	Key	Description
metric_id	INTEGER	Primary Key	Unique metric identifier
metric_name	VARCHAR(100)		Labor market metric name

2. geography

Purpose

Stores geographic area information used for labor market analysis.

Examples:

- Louisville/Jefferson County, KY-IN Metropolitan Area
- Kentucky

Primary Key

geography_id

Relationships

One geography can relate to many series records.

Referenced by:

series.geography_id

Table Structure

Column Name	Data Type	Key	Description
geography_id	INTEGER	Primary Key	Unique geography identifier
area_code	VARCHAR(20)		BLS geographic area code
area_name	VARCHAR(100)		Geographic region name

3. industry

Purpose

Stores labor market industry classifications used for dashboard filtering and analysis.

Examples:

- Manufacturing
- Retail Trade
- Government
- Transportation and Warehousing
- Private Education and Health Services

Primary Key

industry_id

Relationships

One industry can relate to many series records.

Referenced by:

series.industry_id

Table Structure

Column Name	Data Type	Key	Description
industry_id	INTEGER	Primary Key	Unique industry identifier
industry_code	VARCHAR(20)		BLS industry classification code
industry_name	VARCHAR(100)		Industry category name

4. demographics

Purpose

Stores demographic filtering categories for potential future dashboard expansion.

Note: Demographic data is not currently visualized within the MVP dashboard but remains a part of the schema to support future age and gender-based labor market analysis.

Primary Key

demographic_id

Relationships

One demographic category can relate to many series records.

Referenced by:

series.demographic_id

Table Structure

Column Name	Data Type	Key	Description
demographic_id	INTEGER	Primary Key	Unique demographic identifier
gender	VARCHAR(50)		Gender category
age_group	VARCHAR(50)		Age group category

5. series

Purpose

Stores metadata describing each BLS series and links each series to metric, geography, industry, and demographic dimensions.

This table functions as the central dimension table for labor market analytics.

Primary Key

series_id

Foreign Keys

- metric_id → metrics.metric_id
- geography_id → geography.geography_id
- industry_id → industry.industry_id
- demographic_id → demographics.demographic_id

Relationships

- Many series can reference one metric.
- Many series can reference one geography.
- Many series can reference one industry.
- Many series can reference one demographic category.
- One series can relate to many observations.

Table Structure

Column Name	Data Type	Key	Description
series_id	VARCHAR(50)	Primary Key	Unique BLS series identifier
metric_id	INTEGER	Foreign Key	References labor market metric
geography_id	INTEGER	Foreign Key	References geographic region

industry_id	INTEGER	Foreign Key	References industry category
demographic_id	INTEGER	Foreign Key	References demographic category

6. observations

Purpose

Stores monthly labor market observations retrieved from the BLS API.

Observation data includes:

- Unemployment rates
- Employment levels
- Labor force values
- Industry employment values

Composite Primary Key

(series_id, year, period)

Foreign Keys

series_id → series.series_id

Relationships

Many observations can reference one series.

Each observation is uniquely identified by:

- series_id
- year
- period

Table Structure

Column Name	Data Type	Key	Description
series_id	VARCHAR(50)	Composite PK / FK	References BLS series
year	INTEGER	Composite PK	Observation year
period	VARCHAR(10)	Composite PK	Monthly period code
value	NUMERIC		Labor market metric value

Cardinality Relationships

Parent Table	Child Table	Relationship Type
metrics	series	One-to-Many
geography	series	One-to-Many
industry	series	One-to-Many
demographics	series	One-to-Many
series	observations	One-to-Many

Data Sources

Labor market data is sourced from the U.S. Bureau of Labor Statistics Public API V2.

The API utilizes:

Kentucky Labor Market Indicators

- Unemployment Rate
- Employment Level
- Labor Force

Louisville Metropolitan Industry Employment

- Manufacturing
- Private Education and Health Services
- Transportation and Warehousing
- Retail Trade
- Government

PostgreSQL Load Script

```
CREATE TABLE metrics (  
    metric_id SERIAL PRIMARY KEY,  
    metric_name VARCHAR(100)  
);
```

```
CREATE TABLE geography (  
    geography_id SERIAL PRIMARY KEY,  
    area_code VARCHAR(20),  
    area_name VARCHAR(100)  
);
```

```
CREATE TABLE industry (  
    industry_id SERIAL PRIMARY KEY,  
    industry_code VARCHAR(20),  
    industry_name VARCHAR(100)  
);
```

```
CREATE TABLE demographics (  
    demographic_id SERIAL PRIMARY KEY,  
    gender VARCHAR(50),  
    age_group VARCHAR(50)  
);
```

```
CREATE TABLE series (  
    series_id VARCHAR(50) PRIMARY KEY,  
    metric_id INTEGER,  
    geography_id INTEGER,  
    industry_id INTEGER,
```

```

    demographic_id INTEGER,

    CONSTRAINT fk_metric
        FOREIGN KEY (metric_id)
        REFERENCES metrics(metric_id),

    CONSTRAINT fk_geography
        FOREIGN KEY (geography_id)
        REFERENCES geography(geography_id),

    CONSTRAINT fk_industry
        FOREIGN KEY (industry_id)
        REFERENCES industry(industry_id),

    CONSTRAINT fk_demographics
        FOREIGN KEY (demographic_id)
        REFERENCES demographics(demographic_id)
);

CREATE TABLE observations (
    series_id VARCHAR(50),
    year INTEGER,
    period VARCHAR(10),
    value NUMERIC,

    PRIMARY KEY (series_id, year, period),

    CONSTRAINT fk_series
        FOREIGN KEY (series_id)
        REFERENCES series(series_id)
);

```

SQLAlchemy Database Integration (using Python)

SQLAlchemy was used to establish a connection between Python and the PostgreSQL database. Pandas DataFrames containing labor market data retrieved from the BLS API were loaded into PostgreSQL tables using SQLAlchemy's `create_engine()` connection interface and the `to_sql()` method as part of the ETL pipeline.

```

from sqlalchemy import create_engine

engine = create_engine(
    "postgresql+psycopg2://postgres:password@localhost:5432/bls_db"
)

```

```
df_series.to_sql(  
    "series",  
    engine,  
    if_exists="append",  
    index=False  
)
```

```
df_observations.to_sql(  
    "observations",  
    engine,  
    if_exists="append",  
    index=False  
)
```

Incremental Loading Strategy

Duplicate observation records are prevented through:

- Composite primary key structure
- Duplicate record validation
- Removal of duplicate records using:

```
df_observations.drop_duplicates(  
    subset=["series_id", "year", "period"]  
)
```

This ensure that monthly labor market observations are loaded only once into the database.